Vector Magnitude & Direction: Rectangular to Polar Coordinates

Right Triangle Trigonometry & Vectors • Standard: CCSS.MATH.HSN.VM.A.1

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1. MATHEMATICAL CONCEPT & STACK APPROACH: Decompose 2D vectors into horizontal and vertical components (dx, dy) and calculate polar magnitude r and direction angle ϑ.

Every 2D vector or right triangle relates two perpendicular legs (dx, dy) to a hypotenuse r and angle ϑ via $r = \sqrt{(x^2 + y^2)}$ and $\tan(\vartheta) = y/x$. Keeping horizontal component dx in Register Y and vertical component dy in Register X pairs the two coordinates naturally.

2. MODELING WITH BLOCKS (WHEN APPLICABLE) & STUDENT PREDICTION

Published Block Kit: Right-Triangle & 2D Vector Component Diagram

Hands-On Setup: Draw a right triangle with horizontal run dx = 4 (Register Y) and vertical rise dy = 3 (Register X) to model the 3-4-5 vector.

Student Prediction (Before Calculator): Run = 4, Rise = 3. Predict magnitude: $r = \sqrt{(4^2 + 3^2)} = 5$. Predict angle: $\arctan(3/4) \approx 36.87°$.

3. CALCULATOR EXECUTION (SC-32 KEYCAPS & 4-LEVEL STACK TRACE)

Keystroke Flow: [4] [ENTER] [3] [↶] [▸θ,r] [x⇄y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸θ,r]	36.87	Rectangular to Polar! Screen displays angle ϑ = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

4. MATHEMATICAL CONTEXT & REAL-WORLD MOTIVATION (A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow)

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a deep well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure right-triangle ratios! That same relationship between perpendicular components (dx, dy) and polar angle/distance (r, ϑ) guides modern navigation and engineering.

Core Investigation (Student Warm-Up A):

A surveyor walks 60 m East (dx = 60) and 80 m North (dy = 80). Use the Pythagorean theorem and tangent ratio to find the straight-line distance r and angle ϑ north of east.

2D Vector Addition: Resultant Forces by Component Summation

Vector Addition & Resultant Forces • Standard: CCSS.MATH.HSN.VM.A.3

1. FOLLOW-UP EXTENSION CONCEPT, BLOCKS & PREDICTION

Concept & Stack Approach: Non-collinear vectors cannot be added by simply summing their lengths; their horizontal (x) and vertical (y) components must be summed separately first. Summing the x-components and y-components from the inside out gives the net resultant vector $\langle x_{net}, y_{net} \rangle$.

Blocks & Prediction: Sketch component vectors F1 = <3, 2> and F2 = <1, 5> tip-to-tail on a coordinate grid. Prediction: Add components: $x_{net} = 3 + 1 = 4$; $y_{net} = 2 + 5 = 7$. Magnitude $r = \sqrt{(4^2 + 7^2)} = \sqrt{(65)} \approx 8.06$.

2. EXTENSION CALCULATOR SEQUENCE & STACK TRACE

[3] [ENTER] [1] [+] [2] [ENTER] [5] [+] [↶] [▸θ,r] [x ⇄ y]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[3]	3	Horizontal component x1 = 3
[ENTER]	3	Lifted to Y
[1]	1	Horizontal component x2 = 1
[+]	4	Net horizontal force $x_{net} = 4$ in X
[2]	2	Vertical component y1 = 2 (4 lifts to Y)
[ENTER]	2	2 lifted to Y; 4 in Z
[5]	5	Vertical component y2 = 5 in X
[+]	7	Net vertical force $y_{net} = 7$ in X; $x_{net} = 4$ in Y
[↶] [▸θ,r]	60.26	Resultant angle ϑ = 60.26°
[x ⇄ y]	8.06	Resultant force magnitude r = 8.06!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Evaluating trigonometric functions in Radian mode when angles are given in Degrees (or vice versa).

Hands-On Block Remediation: Have students check the DEG/RAD indicator on the SC-32 status bar and verify with the 1/2 (42 mm) and 1 Whole (84 mm) Fraction Studio towers that sin(30°) = 42/84 = 0.5000 before starting vector problems.

Misconception 2: Swapping x and y components when converting rectangular coordinates to polar via [→ϑ,r].

Hands-On Block Remediation: Place horizontal Run (x) in Row 1 (Register Y) and vertical Rise (y) in Row 0 (Register X) of the Place Value Tray before pressing [→ϑ,r].

Extension Challenge (Student Warm-Up B):

Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector $\langle x_{net}, y_{net} \rangle$ and its magnitude r.

Expected Answer: Resultant <3, 6>; Magnitude $r = \sqrt{(3^2 + 6^2)} = \sqrt{45} \approx 6.71$.

Trigonometry, Vectors & Polar/Rectangular Coordinates

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: RIGHT-TRIANGLE RATIOS & 2D VECTOR DECOMPOSITION RULES

- Trigonometric functions are ratios of right-triangle sides: $\sin(\vartheta)$ = Opposite / Hypotenuse, $\cos(\vartheta)$ = Adjacent / Hypotenuse, $\tan(\vartheta)$ = Opposite / Adjacent.
- Every 2D vector v = <dx, dy> forms a right triangle with magnitude $r = \sqrt{(dx^2 + dy^2)}$ and direction angle $\vartheta = \arctan(dy / dx)$.
- To add two vectors A = <x1, y1> and B = <x2, y2>, add their horizontal components (x1 + x2) and vertical components (y1 + y2) separately.

Key Sequence: 4 ENTER 3 ↶ ▸θ,r x ≂ y

Investigation 1: Find the magnitude r and direction angle ϑ for the vector <12, 5>.

1. Right Triangle / Vector Sketch:

2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 2: Find the hypotenuse of a right triangle with legs of length 7 and 24.

1. Right Triangle / Vector Sketch:

2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 3: In a 30°-60°-90° triangle, the side opposite 30° is half the hypotenuse (42 mm vs. 84 mm). What is sin(30°)?

1. Right Triangle / Vector Sketch:

2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Investigation 4: A vector has magnitude r = 10 at angle ϑ = 45°. Find its horizontal ($x = r \cos \vartheta$) and vertical ($y = r \sin \vartheta$) components.

1. Right Triangle / Vector Sketch:

2. Geometric / Trig Formula Steps:

Formula / Trig Ratio:

Substitute & Square / Eval:

Root / Angle Solution:

3. My Prediction:

Final Verified Answer:

[ ] Checked on SC-32

Unit 09: Trigonometry, Vectors & Polar/Rectangular Coordinates — Answer Key

Complete solutions and SC-32 verification keystrokes for Grades 9–11

1. CORE (WARM-UP A) & EXTENSION (WARM-UP B) SOLUTIONS

Core Investigation (Warm-Up A): A surveyor walks 60 m East (dx = 60) and 80 m North (dy = 80). Use the Pythagorean theorem and tangent ratio to find the straight-line distance r and angle ϑ north of east.

Core Solution: $r = \sqrt{(60^2 + 80^2)} = 100$ m; $\vartheta = \arctan(80/60) \approx 53.13°$.

Core Check Keys: 6 0 ENTER 8 0 ↶ ▸θ,r $x \rightleftarrows y$

Extension Challenge (Warm-Up B): Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector $<x_{net}, y_{net}>$ and its magnitude r.

Extension Solution: Resultant <3, 6>; Magnitude $r = \sqrt{(3^2 + 6^2)} = \sqrt{45} \approx 6.71$.

Ext. Check Keys: 5 ENTER 2 +/− + 2 ENTER 4 + ↶ ▸θ,r $x \rightleftarrows y$

2. STUDENT LAB WORKSHEET SOLUTIONS (INVESTIGATIONS 1–4)

Investigation 1: Find the magnitude r and direction angle ϑ for the vector <12, 5>.

Expected Answer: r = 13, ϑ = 22.62°

SC-32 Verification Keys: 1 2 ENTER 5 ↶ ▸θ,r

Investigation 2: Find the hypotenuse of a right triangle with legs of length 7 and 24.

Expected Answer: $r = \sqrt{(49 + 576)} = 25$

SC-32 Verification Keys: 7 ENTER 2 4 ↶ ▸θ,r $x \rightleftarrows y$

Investigation 3: In a 30°-60°-90° triangle, the side opposite 30° is half the hypotenuse (42 mm vs. 84 mm). What is sin(30°)?

Expected Answer: 42 / 84 = 0.5

SC-32 Verification Keys: 3 0 SIN

Investigation 4: A vector has magnitude r = 10 at angle ϑ = 45°. Find its horizontal (x = r cos ϑ) and vertical (y = r sin ϑ) components.

Expected Answer: x ≈ 7.07, y ≈ 7.07

SC-32 Verification Keys: 1 0 ENTER 4 5 ↷ ▸y,x

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.